

EMERGING TECHNOLOGIES OVERVIEW AND APPLICATIONS

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DCIT 26 (Application Development and Emerging Technologies)

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1.0 Executive Summary

This report aims to present a brief description of what emerging technologies are, their definition, key characteristics, impact on technology-related professions, and effects on society, based on the information provided in Module 1 lecture notes. In addition, this report will also discuss the basic principles and characteristics of technologies such as augmented reality (AR), virtual reality (VR), and artificial intelligence. It is important for students and professionals to understand the characteristics and effects of these new and innovative developments in order to further excel in the dynamic business world.

2.0 Definition and Key Characteristics of Emerging Technologies

2.1. Definition of Emerging Technologies

In general, emerging technology refers to a technological field that is constantly evolving and has the potential to fundamentally change the way business is done. Although many people tend to mistake “new” or “recently innovated” technologies as “emerging,” this assumption is far from the truth. One of the common examples of emerging technology is artificial intelligence (AI), which has been in existence since the 1950s but continues to be expanded, developed, and

applied in different fields even to this day.

2.2 Five Core Characteristics

Based on the baseline criteria, a technology can be identified as emerging if it possesses five core characteristics namely:

- **Radical Novelty:** This suggests that it introduces novel paradigms rather than iterative improvements to existing systems.
- **Relatively Fast Growth:** It experiences rapid adoption, investment, and development networks expansion.
- **Convergence:** This technology converges with and complements existing or related technologies.
- **Prominent Impact:** This refers to its ability to drive significant changes in several industries rather than single niche markets.
- **Uncertainty and Ambiguity:** This technology operates in a dynamic environment, and the ultimate market outcomes and implications are unclear.

3.0 Importance of Emerging Technologies for IT and CS Professionals

3.1 Industry Relevance and Tool Competency

The understanding of this concept by IT and Computer Science professionals is important since these solutions present the fundamental tools and systems used in the current industry. Without sustained competence in key emerging fields such as Artificial Intelligence, computer science, and IT, practitioners will find it difficult to remain relevant as business organizations embrace these systems at an increased rate.

3.2 Career Roadmap and Technical Analysis

When viewed from an analytical viewpoint, this knowledge offers a roadmap into the technology field for these experts. They can use this information to understand downstream engineering concepts and the market dynamics that shape the emerging capabilities landscape. It also aids them in assessing the vendor landscape and exploring career opportunities made possible by the new tools and systems.

4.0 Societal Impact of Emerging Technologies

4.1 Positive Impacts

- **Operational Efficiency and Healthcare Advancements:** High-performance computing and artificial intelligence play a significant role in enhancing the drug discovery process and diagnostic testing, thus accelerating healthcare delivery.
- **Seamless Connectivity:** The Internet of Things (IoT) facilitates effective internet connectivity, communication, and control of physical objects, enabling automated operations.

4.2 Negative & Unintended Impacts

- **Long-Term Uncertainty:** Its rapid adoption occurs at a faster rate than its ultimate implications, and risks are understood.
- **Ethical and Regulatory Friction:** The varying viewpoints on how this should be regulated cause friction between different stakeholders, resulting in unnecessary delays.

5.0 Comparative Analysis: Augmented Reality (AR) vs. Virtual Reality (VR)

Although both Augmented Reality (AR) and Virtual Reality (VR) are types of Industry 4.0 immersive technologies, there are some significant differences between them, which are explained below:

5.1 Augmented Reality (AR): This technology allows computer-generated elements (3D models, graphics, and information) to be overlaid into the existing physical environment. In this case, users are aware of their real-world surroundings while interacting with augmented elements.(e.g., retail furniture visualization applications).

5.2 Virtual Reality (VR): Completely replaces the physical environment with a fully simulated digital world. VR isolates the user from real-world surroundings using head-mounted displays, delivering an end-to-end immersive experience designed for deep simulation, training, or virtual collaboration.

6.0 Strategic Analysis of Artificial Intelligence

6.1 Advantages (Pros) of AI

- **Processes Large Volumes of Data Quickly:** Artificial Intelligence has the ability to process large amounts of information in a short time. This makes it possible to analyze thousands or millions of patients' scans and test results within a short time, thus facilitating quick diagnosis.
- **Cross-Industry Application:** Artificial Intelligence can be leveraged in different fields, including medicine, marketing, finance and banking, and automated transportation. •

Increases Operational Efficiency: Artificial Intelligence can be integrated into systems and processes to enhance efficiency. Unlike manual automated processes, AI can complete one task after another, thus increasing productivity.

- **Continuous Technological Evolution:** The technology space in which AI resides is evolving at a very fast rate, thus making the various capabilities and applications evolve at a faster rate.
- **Enables Novel Problem-Solving:** One of the key advantages of using this technology is that it can help identify non-linear relationships in data sets, thus leading to the discovery of unique solutions to problems.

6.2 Disadvantages and Risks (Cons) of AI

- **Algorithmic Bias and Data Quality Risks:** Since AI is dependent on data, any form of prejudice or weaknesses in data will be reflected in its results or decisions made.
- **Job Displacement and Workforce Transition:** One of the key risks of adopting this technology is the fact that it can replace most if not all jobs, creating great social impact and putting all individuals at risk of losing their jobs.
- **Unintended Consequences and Hallucinations:** AI can exhibit unintended behaviors or consequences, thus presenting great risks to the business ecosystem.
- **Ethical, Regulatory, and Governance Challenges:** The differing viewpoints regarding this technology and how it should be governed or protected present a key challenge in its adoption, especially in the development of protective strategies.
- **Security Vulnerabilities:** The adoption of this technology comes along with some cybersecurity risks, including prompt injection and data poisoning, that require the need to constantly update the tools, models, and overall security measures.

7.0 References

Module 1 Lecture: Application Development and Emerging Technologies (DCIT 26).
Department of Information Technology, Cavite State University - Silang Campus.

8.0 Appendix

Appendix A: Verification of Originality and Writing Performance

Figure 1: GPTZero AI Detection Results (GPTZero - 100% Human)

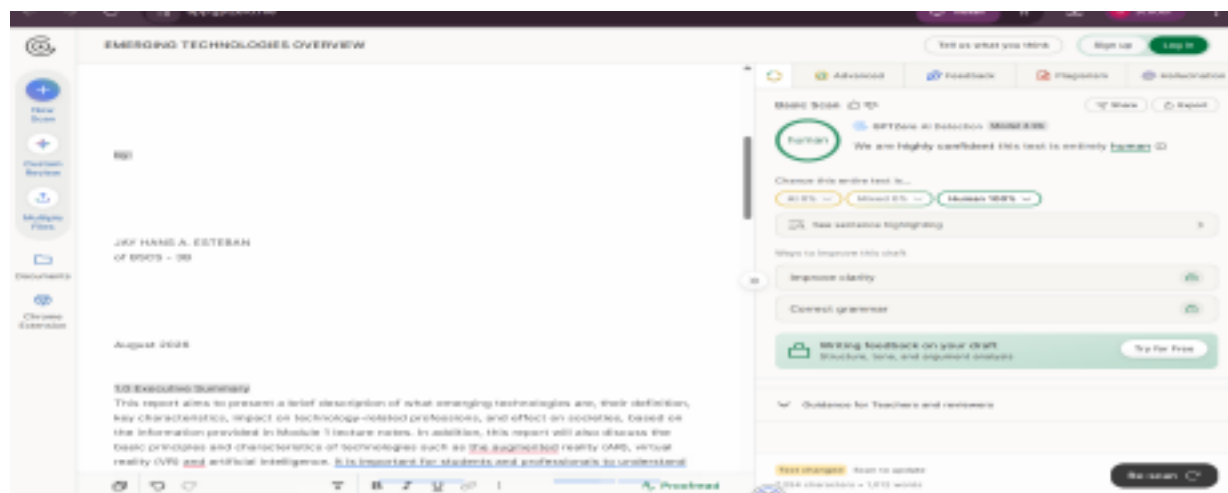


Figure 2: Grammar and Writing Performance Check (QuillBot Grammar Checker)

